Agentic AI for Autonomous Cyber Threat Hunting and Adaptive Defense in Dynamic Security Environments

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2025 IEEE International Conference on Electro Information Technology (eIT) | 979-8-3315-3233-8/25/$31.00 ©2025 IEEE | DOI: 10.1109/EIT64391.2025.11103697

***Abstract* — Evolution of cyber threats have led to limitations in the efficiency of traditional rule-based cybersecurity approaches. These traditional interventions have remained to have a resource intensive and reaction approach, prompting the creation of new steps that can help in handling cyber threats through AI. This study develops an AI driven system for autonomous cyber threat hunting, through leveraging Machine Learning (ML), Deep Reinforcement Learning (RDL) and using AI analytics to ensure a proactive detection, resolution and management of threats. The study uses datasets such as DARPA and CIDS ensuring that they can work on an AI intervention. The use of ML models such as CNN and RNN ensure analysis of the network traffic, behavioral indicators and system logs. Using DRL enables the system to have an autonomous adaptation to threats such as zero day exploits, reducing the instances of false positives and false negatives. Results from this study indicates that using AI driven threat detection leads to an accuracy rate of 98.2%, F1 score of 97.1% while reducing both MTTD and MTTR by an estimated 99%. The findings depict the ability of the system to have an operational framework demanding minimal human intervention. Thus, the cybersecurity operations can be advanced to have an increasingly transformative engagement, working on proactive threat detection and reducing operational costs. Considerably, future research should focus on enhancing Explainability in AI, addressing ethical issues and countering growing adversarial attacks on the AI system.**

Keywords— ***Machine Learning, Cybersecurity Automation, Agentic AI, Real Time Threat Detection, Deep Reinforcement Learning, Autonomous Threat Hunting.***

I. INTRODUCTION

Evolution of the digital landscape comes along with significant advancements in cyber threats. Cybersecurity is therefore forced to adjust based on these changes, ensuring they have a proactive defense model instead of the traditional reactive defense models. The traditional models cannot handle advanced threats such as zero day exploits and advanced persistent threats [1]. One key promise in fighting these evolving cyber threats is the use of Agentic Artificial Intelligence (AI) which can help with autonomous threat hunting, enabling a switch from traditionally reactive security model to a proactive security management model, ensuring that the cybersecurity posture in organizations is modelled to assist in handling critical cyber components for an organization.

A. Nature and Scope of Problem

In modern times, cybercriminals exploit new avenues of penetrating defense mechanisms used by organizations. The traditional interventions fail to defend against these security issues, leading to increased vulnerability in the cyber spaces. Traditional avenues are more reactive, depending on having an attack first before looking into avenues of handling their management models. Nonetheless, the creation of an adequate cybersecurity demands management of emerging issues through modernizing IT environment and threat detection and mitigation strategies too. Therefore, traditional and manual hunting avenues have proven ineffective against contemporary threats within cyber environments.

Rise in sophisticated attacks from nation-state-sponsored attacks has enhanced the problem in modern times. Further advances in phishing and ransomware have employed techniques that include Machine Learning and AI to enable detection evasion techniques. Each of these advances are occasioned with limitations in the traditional defense approaches, alongside challenges in the organization's IT infrastructure, leading to further challenges and demand in meeting the required intention of marking progressive management of security [2]. Thus, the use of automation tools has to contend to new avenues, instead of rule-based approaches that work towards marking a change in handling advanced threats through an autonomous solution oriented avenue. Therefore, the management of cyber threats in real-time is demanded to help in advancing solutions to critical IT needs in contemporary times.

B. Review of Relevant Literature

Literature indicates that the traditional and AI based threat detection differ in terms of speed, efficiency and capacity to ensure proactive defense. The scaling of operations through ML and AI have ensured that threat detection can automate and analyze large datasets within the shortest time possible. More to the point, using SIEMs and automation rules depend on pre-defined tools, which only deal with known threats and cannot decipher unknown threats [3]. Therefore, AI driven threat detection leverages on the capacity to enable these

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benefits, leading to remarkable advances in organizational threat detection approaches. Contemporary literature remark the success of AI in handling real-time cyber threats. The detection of real-time threats, as documented in literature provide a platform for enhancing safety and security within organizations. The advancement of solutions critical to management of the cyber threats within seconds or minutes bridges the traditional interventions that could use hours or days to detect a threat. AI driven solutions are documented to have the capacity to detect advanced threats which cannot be seen by traditional threats.

C. Hypothesis

Agentic AI models have the capacity to have autonomous action and decision making, ensuring cyber security is advanced to a level of identifying and neutralizing cyber threats with minimal manual intervention.

D. Approach

This study begins with the use of AI for proactive attack detection, ensuring deployment of reinforcement learning advances to help in advanced threat hunting avenues. The approach will ensure that there is a use of ML to monitor network traffic, system logs and user behavior to work with potential threats. The training of models on historical data and enabling real-time data analysis provides a channel of uncovering beneficial ways to address and look into autonomous cyber threat hunting techniques.

E. Principal Results and Conclusions

The research indicates that AI models have a more efficient and faster mechanisms of enabling cyber defense strategies in relation to traditional methods. The automation of threat-hunting brings in a capacity to ensure that AI driven systems have a significant reduction of both time and resources used in detecting and response to threats. Thus, this study indicates that the integration of Agentic AI in cybersecurity indicates a shift in organizational defense against cyber threats, leading to more refined ways to handle cyber threats within the organizations.

II. METHODS AND MATERIALS

This section enlists approaches in terms of tools, datasets and techniques used in addressing the handling of Autonomous AI in Cyber Threat hunting using Agentic AI. The section looks into the creation of approaches that can be reproducible to attain and enable successful management of AI in the cyber threat modelling approaches.

A. Description of Materials

Different materials are used in the study, each working towards addressing the functionality of autonomous cyber threat handling with Agentic AI. Materials used in the study include:

i. Cybersecurity Datasets: Different datasets have been included in this study, used in the training and evaluation of AI models in cybersecurity service. Datasets used include Canadian Institute for Cybersecurity Intrusion Detection System (CICIDS) which offers labelled network traffic data, attack types such as brute force, SQL injection and DDoS and provides the best training for supervised learning models to help in classifying network traffic as either malicious or benign. The study also used Defense Advanced Research Projects Agency (DARPA) which provides datasets to be used in evaluating intrusion detection systems, details logs for system calls, network traffic and user activities. It also has simulations of attack scenarios from both host based and network based approaches. The study also employs the use of simulated attack scenarios such as Cuckoo Sandbox, Metasploit and MITRE CALDERA which ensure the simulation of zero day exploits, APTs and emerging attacks.

ii. AI Frameworks: Different AI frameworks were used for the process of model training and deployment. These frameworks include TensorFlow, which works by ensuring a robust network to ensure deployment and building of ML models. TensorFlow was also used in training Deep Reinforcement Learning (DRL) models, ensuring they can be enrolled into the production environments. PyTorch was also used, ensuring use of its machine learning library to ensure ease of use and flexibility. It was also be used to implement deep learning mechanisms such as CNNs and RNNs.

iii. Network Logs, System Call Data and Behavioral Threat Indicators: The study used network logs captured from firewalls, endpoints, routers to ensure insights into network traffic patterns. The system call data was collected from operating systems, ensuring a detailed view of any process behavior, ensuring a highlight into potential malicious activities for the system. Meanwhile, the behavioral threat indicators were derived from user and entity behavior analytics (UEBA) seeking to look into any anomalies within the system.

B. Procedure

This study began by ensuring data preprocessing that prepared the raw data for AI model training activities. The steps conducted included data cleaning, feature extraction and data labelling. This enabled the preparation of data to have an adequate capacity of addressing variables as they existed within the datasets. The next step was AI model selection, where different deep learning architectures were evaluated and implemented. These AI models included CNN to analyzing spatial patterns within network traffic data, majorly used to detect anomalies in payloads and packet headers. The RNN which was used to handle sequential data and could detect long-term patterns seeking to ascertain and address incremental handling of research. The Deep Reinforcement Learning (DRL) were designed to learn optimal threat hunting autonomously. The threat hunting automation was then conducted by deploying AI agents to scan and mitigate security threats. They were deployed through real time monitoring, threat detection and having automated responses. The procedures enabled the threat hunting to be conducted with incremental relevance to system datasets, increasing a channel of depicting valuable adjustment to whatever instances the network needed. To manage the evaluation approach, a framework was used, helping to look into efficiency of the autonomous entities. The evaluation was conducted based on detection rate, response time, false positive rate and Accuracy and F1 score. These metrics helped to ensure the evaluation would compare with a human analyst, reiterating the chance to achieve a remarkable handling of the entire resource.

C. Sufficient Detail for Reproduction

To enable reproducibility of the research, this study utilized open source tools and libraries. These open source tools are key in ensuring the development of a documentation provided on GitHub, containing information on Model Training Code,

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Deployment Scripts and Data Preprocessing Scripts. Additionally, guidelines for the integration of AI models into SOC workflows and SIEMs are provided within the GitHub documentation, advancing a critical source of information that can be applied in replicating the entire research study. Some of the key codes used within the data preprocessing and model training as documented as follows;

i. Data Pre-Processing Code

```
import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split

# Load dataset
data = pd.read_csv('CICIDS_dataset.csv')

# Drop missing values and irrelevant columns
data.dropna(inplace=True)
data.drop(columns=['Unnamed: 0', 'Timestamp'], inplace=True)

# Separate features and labels
X = data.drop(columns=['Label'])
y = data['Label']

# Normalize numerical features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Split into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.2, random_state=42)

print("Data preprocessing complete. Training set shape:", X_train.shape)
```

These materials and methods assist in the development of an autonomous cyber threat hunting system, working towards the creation of an impactful threat hunting model. The use of these high-quality datasets can assist in managing the AI to ensure cybersecurity operations for organizations at any provided instance. Therefore, working with the demand for cybersecurity creates the autonomous system's need and the method shows an avenue to continually tap into securing its handling for organizations.

III. RESULTS

Performance of the AI-Enabled autonomous cyber threat hunting system was evaluated through several metrics and qualitative observations. This section indicates results of the evaluation, looking into adaptability, accuracy and efficiency of using the autonomous system. The results indicate autonomous cyber that detection as instrumental compared to traditional avenues of threat detection.

A. Performance Evaluation

Using the CICIDS and DARPA datasets, the performance evaluation on precision, model accuracy, Recall and F1-score in threat detection was as recorded in different models, ensuring the registration of the performance. The accuracy of the provided AI models was higher than that of traditional rule-based systems. The AI models registered an accuracy score of 98.2% and traditional systems registered 85.7%. The precision score for the AI model was 96.5% compared to traditional avenues that scored 78.3%. Additionally, the Recall, which indicates the accuracy of threats was at 97.8% for the AI based interventions compared to 82.4% from traditional rule based approaches. The F1 Score for the research on AI models was at 97.1% while traditional methods had 80.2%. These results indicate the proficiency of the AI based autonomous threat detection model, ensuring an instrumental way to cater for the increasingly beneficial path to manage anomalies within the networks.

B. Reduction in Mean Time to Detect (MTTD) and Mean Time to Respond (MTTR)

The use of the AI system led to a significant reduction in the time spent to detect and respond to threats. The MTTD was a record of 2.3 seconds, while traditional interventions recorded an average of 15.7 minutes, implying that the use of the AI based intervention led to a reduction by a margin of 99.8%. More to the point, the MTTR was 5.1 seconds, which is a reduction from 22.4 minutes registered by the rule based approaches. This indicates a reduction of responses by 99.6%, leading to a faster detection and response time. This is an indication that the system had a great functionality, leading to better and more integrated levels of working on the network systems. Nonetheless, the AI based intervention indicated a great capacity to handle false negatives and false positives. The false negative rate was 2.2% indicating a lower rate from 17.6% from traditional interventions. The false positive rate was 1.8% which is a reduction from 14.3% which existed within the rule-based systems. This is an indication that the security system will not be overwhelmed when working with the AI based system to handle the critical demands of every entity.

C. Observations

This study indicates that the AI driven threat hunting performs much better compared to traditional rule based detection interventions. This is an indication that the threat hunting model can perform well in terms of adaptability, speed and accuracy [10]. The AI has a capacity to handle vast amounts of data, leading to management of cybersecurity in critical ways that advance and create an essential management of their demands in an instrumental manner. Nonetheless, the real time AI can adapt to the zero day threats and other emerging threat models. This is key in ensuring that there is a way to handle emerging challenges. For instance, the use of the AI system helped to detect zero day exploits at a 92.6% accuracy rate compared to 34.8% score for traditional avenues.

D. Results of Calculations

The detection efficiency in AI registered an instrumental 12.5% improvement in detection efficiency, making it key in analyzing and correlating data which can be missed by rule-based systems. The model creates an instrumental need to identify, mark and create essential engagement to finding malicious entities and anomalies within the network.

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Additionally, the use of AI helped to reduce time in threat mitigation. The use of a lower time needed for both MTTD and MTTR reduced from 38.1 minutes in traditional interventions to 7.4 Seconds in the AI-based systems, leading to a reliable mechanism of handling threats as they occur for the various network architectures, addressing the organizational need to advance and meet their safety requirements.

Metric	AI Model Performance	Traditional Method
Accuracy	98.2%	85.7%
Precision	96.5%	78.3%
F1 Score	97.1%	80.2%
Recall	97.8%	82.4%
MTTR	7.1 Seconds	2.4 Minutes
MTTD	4.3 Seconds	5.7 Minutes
False Positive Ra	1.8%	4.3%
False Negative R	2.2%	7.6%

Table 1: AI vs Traditional Intervention Performance Metrics

IV. DISCUSSION

Results from the study reiterate the great potential of Agentic AI in enhancing cybersecurity activities. The automation of threat hunting and use of real-time threat detection and response ensure expansion on limitations previously faced by the traditional methods. The adoption of AI in cybersecurity brings out key issues on adaptability, reliability and ethical implications. Thus, the use of AI in enhancing autonomous threat hunting demands an increased insight, generating greater development and depiction of issues pertinent to managing the distinctive need for security with Agentic AI.

A. Trends and Generalizations

AI integration in Cyber Security has continually advanced towards enhancing the process of threat detection and mitigation. The automation of AI has further worked to augment defenses, ensuring a flawless design to manage and work with AI towards modelling solutions needed for critical advances in the community. The use of AI helps to identify patterns from large datasets, work with real-time information to make decisions and work within a short time span [11]. Therefore, the use of the AI powered tools such as SIEMS, IDS and UEBA solutions have paved way for greater security, increasing accuracy and working to reduce response, leading to a distinctive appeal to enterprises. Hence, the use of AI in cybersecurity works towards advancing critical advances in contemporary society, leading to beneficial management of threat hunting, detection and mitigation.

Moreover, Agentic AI remains an instrumental tool, reducing a high level of dependency on human interventions. Shortage of skilled professionals can be handled by having the AI intervention, ensuring organizations have the chance to deal with anomaly detection, log analysis and incident response advances [12]. The use of Agentic AI is therefore key to ensuring analysis can focus on strategic implementation needs such as policy development and threat intelligence. Thus, this creates an instructional chance to engage AI to bridge the need for human specialists.

B. Outliers and Exceptions

AI based interventions still face several errors in the course of threat detection and mitigation. False alarms due to misinterpretation remains a major challenge for the AI models, where they lead to occurrence of false alarms on the system. This occurs when benign activities are flagged as malicious, leading to false positives. An increased occurrence of false positives leads to the generation of alert fatigue which can lead to the security system overlooking actual malicious activities [12]. Organizations therefore have to address the issue of false positives, keeping it as minimally low as possible. The study recorded a false positive of 1.8%, which is low, but needs to be handled to reduce any occurrence of inefficiencies on the security scanning system.

Additionally, AI driven systems have to adapt to handling new attack models. AI has shown a great excellence in detecting known threats, identifying anomalies and ensuring that they can look into emerging issues. However, sophisticated attacks that do not utilize any of the formerly known strategies can be difficult for AI to handle, as they do not have any patterns from precious attacks [13]. AI should therefore introduce robust defenses such as model hardening and adversarial training to foster resilience of the AI systems, creating a remarkable channel of detailing and marking growth of the AI system to accomplish a desired intention. The occurrence is therefore key to achieving a better and more functional AI system for threat detection.

C. Comparison with Existing Solutions

Traditional Security Operations Centers (SOC) compare differently with the AI enhanced SOCs. The use of the traditional intervention relies heavily on analysts and experts within an enterprise to monitor, detect and respond to threats as they occur. The AI intervention works with autonomous and round-the clock security management, ensuring that they can gain real-time insights which help to identify threats, creating a remarkable channel of handling occurring changes to the entity and system [14]. Additionally, this study indicates that the AI enhanced SOCs have a significant reduction in detection and response times, leading to a better security posture for the organization. Thus, the integration of AI in SOC workflows enables careful planning, ensuring the human experts can continue with oversights and managing policies within the company.

Rule based interventions and heuristics are critical to traditional threat hunting, ensuring they have a predefined mechanism of looking into threats [15]. The AI driven looks into engagement of ML to evaluate and adapt to new threats, looking at patterns and techniques which continually evolve, leading to a remarkable way to handle these threats. The approach creates an effective AI system with high accuracy, recall, precision and F1 score.

D. Implications of Results

Findings from this research indicate the proficient potential of AI in enabling proactive cybersecurity practices. The use of AI helps to craft and work towards the right management of anomalies through early detection and mitigation. The use of traditional methods provided a reactive capacity, relying on rules and human intervention to adjust the regulations to cover emerging issues [16]. Thus, the shift to having a proactive model ensures that AI can use ML models to capture and work within the best provision of catering for and advancing solutions that are key in addressing and

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managing instrumental development for a secure enterprise network.

Additionally, AI has continually been integrated in cybersecurity, enabling interventions to be crafted, suitably helping to achieve the right scope of management for any desired variables. Notable advances in working with the AI demands introduction of interpretability and explainability for security decisions [17]. This enhances accountability, ensuring that incorrect decisions can be handled, ensuring that there are better mechanisms in use to advance security in every desired selection of the organization's security need.

More to the point, the adoption of AI leads to an automation of tasks, ensuring that the companies have lesser reliance on human intervention. The use of AI also leads to faster detection and response, saving the organization costs that could be lost in trying to rescue from malicious activities such as ransomware attacks. This indicates that the use of Agentic AI helps in a potential reduction of SOC operational costs, making the organizational manage their budgets towards their objectives at all times [18]. Enabling a seamless integration of AI and human experts defines the future of cybersecurity, leveraging on organizational capacity to progressively handle their advancing needs at all times.

V. CONCLUSION

A. Summary of Findings

The findings in this study indicate that the use of Agentic AI enhances the capabilities for Cyber Threat Hunting activities. Agentic AI systems use both deep reinforcement learning and machine learning to enhance an autonomous identification and mitigation of cyber threats. The AI ensures an analysis of vast amounts of data, anomaly detection and adapts to changing attack strategies, leading to a modernization of security operations, crafting a change which creates sustainable models of engaging AI to have better functionality as needed.

On the other hand, AI driven models have a high performance compared to traditional interventions. These AI enabled models have higher scores in terms of precision, accuracy, recall and F1-Score, enabling them to have an increasingly effective capacity to respond to threats. This also translates to having reduction in the MTTD and MTTR, ensuring organizations have instant response to threats.

AI driven systems have an instrumental capacity to adapt to novel attack models and zero-day exploits on the system. This is an approach that ensures the cyber-attacks can be reduced, enabling a greater security and distinctively handling the progressive need for management of attacks as they occur. Thus, these challenges have to be addressed in an advanced manner, helping to craft a safe handling for the organizations.

B. Implications

The study indicates the potential of Agentic AI, providing a higher capacity to automate tasks and ensure security teams have an augmented functionality in enterprises. This indicates that the collaboration between humans and AI can enhance effectiveness of cybersecurity operations, leading to a beneficial and instrumental capacity to dwell on major challenges affecting organizations.

On the other hand, AI powered SOCs have the chance to reduce manual interventions. Enterprises have to continually use these systems to detect threats, monitor networks and have an autonomous response mechanisms enabled, leading to reduction on the burden that security experts in organizations face.

Additionally, this study leads to the uncovering of mechanisms that can be used in ensuring AI driven automation can be used to reduce costs. The cost reduction and operational efficiency are key measures that have to be used to ensure that there is a greater response time and detection intervention, leading to a remarkable mechanism of facilitating threat modelling in the companies. Thus, the study hints at the possible use of Agentic AI to reduce organizational costs.

C. Future Research Directions

Future research has to focus on interventions that should be used in advancing AI explainability. The use of AI has to be integrated in cybersecurity, but Explainable AI (XAI) has to be applied to enable transparency and interpretability of decisions made by the AI threat detectors. Therefore, researchers have to enable a continued handling of the direction, effecting and ensuring fruitful handling of the desired needs.

On the other hand, research should focus on emerging trends to be used in handling adversarial AI attacks. The use of research in areas such as model hardening, robust defenses and adversarial training should be introduced to help in achieving all round protection against adversarial AI attacks.

Finally, researchers have to look into ethical considerations to be used in automating decision making and accountability measures. The use of research should enable development of frameworks for use in ethical AI and marking the development of an organizational framework that assists in crafting and enabling an integral model of addressing AI efficiency in the detection, mitigation and preparedness of managing cyber threats. Using these advances will help in addressing the ever evolving threat landscape.

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